

Date	10 July 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	

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1. INTRODUCTION

1.1 Project Overview

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

This project focuses on analysing the food habits, dietary patterns, and lifestyle behaviours of college students using Tableau. The primary objective is to transform raw survey data into meaningful insights through interactive dashboards and visual analytics that help understand students' eating behaviours and their relationship with health, academic performance, and overall well-being.

The dataset used in this project is the **College Food Choices Dataset** containing **125 student records and 66 attributes** related to dietary habits, health perception, food preferences, exercise frequency, eating behaviour, and demographic information.

The project began with data collection and preprocessing. The dataset was imported into Tableau Public, where the data structure was validated by checking field names, data types, missing values, duplicate records, and categorical values. Since the dataset was already cleaned, only minor preprocessing and field mapping were required before visualization. Several calculated fields were created to improve analysis, including **Health Category**, **Breakfast Consumption**, **Comfort Food Reasons**, **Vitamins Consumption**, and **Greek Food Consumption**. Parameters were also created to enable interactive analysis and dynamic visualizations.

Multiple worksheets were developed to examine different aspects of students' dietary behaviours. These include KPI summary cards, breakfast consumption patterns, eating out versus cooking frequency, health category by gender, vitamins consumption by gender, health versus Greek food consumption, people feeling healthy and life rewarded, weight versus exercise frequency, and comfort food reasons.

Interactive dashboards were created by combining these visualizations into a single analytical interface that enables users to explore dietary patterns and identify significant trends. Story points were also developed to present the analysis in a logical sequence, helping users understand how different dietary habits influence students' health and lifestyle. The analysis revealed several important findings. Students who consume breakfast regularly generally report better health and academic focus. Female students constitute a larger proportion of both healthy and unhealthy groups. Stress and boredom were identified as the most common reasons for comfort food consumption. Students who exercise regularly tend to maintain healthier body weights and report better overall well-being.

Overall, this project demonstrates how Tableau can transform raw dietary survey data into meaningful visual insights. The dashboards support educational institutions, nutritionists, and student wellness programs in making informed decisions that encourage healthier lifestyles among college students.

1.2 Purpose

The primary purpose of this project is to analyse the dietary habits, nutritional behaviour, and lifestyle patterns of college students using Tableau dashboards and interactive visualizations.

The project aims to identify relationships between food choices, breakfast habits, exercise frequency, comfort food consumption, health perception, and academic performance. By

presenting these relationships through interactive dashboards, users can better understand factors that influence healthy eating behaviour.

The project also assists universities, nutrition experts, and student wellness coordinators in identifying unhealthy dietary trends and developing strategies that encourage healthier food choices among students.

Through data visualization techniques, the project converts complex survey data into meaningful insights that support evidence-based decision-making for improving student health and academic success.

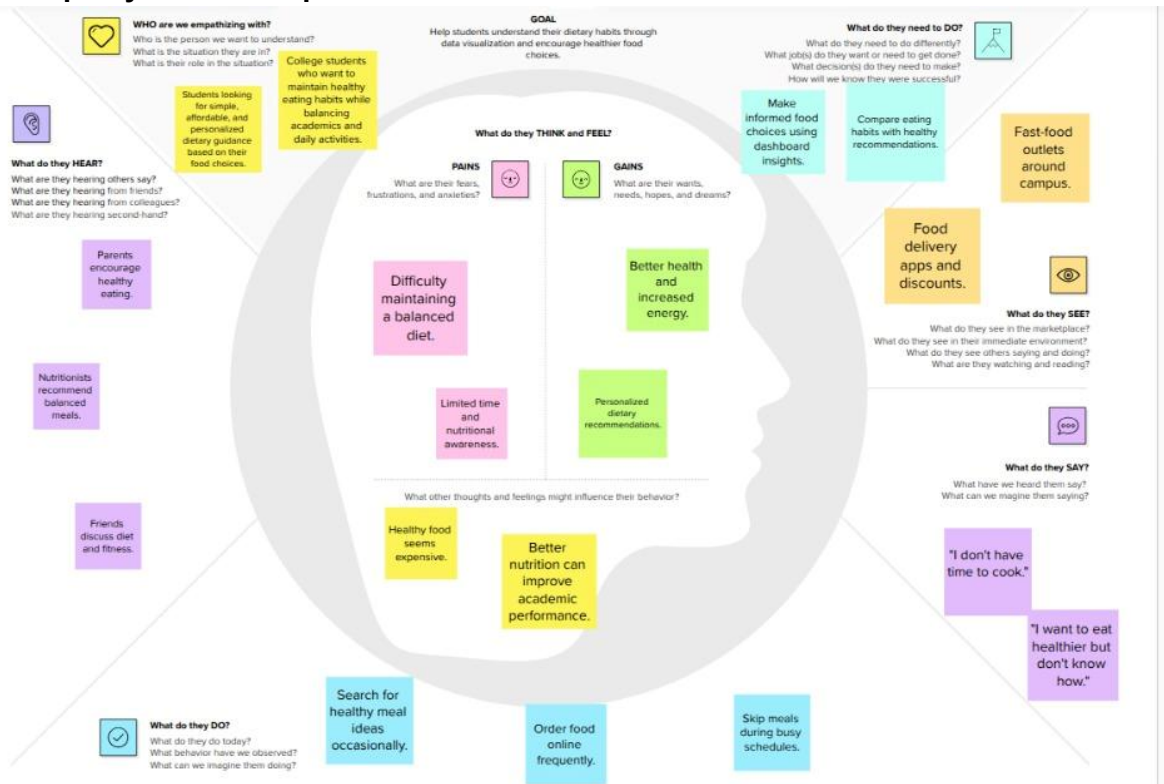
2. IDEATION PHASE

2.1 Problem statement

I am a college student who wants to maintain a healthy diet.	I'm trying to understand my eating habits and choose nutritious food options.	But it is difficult to identify healthy food choices and maintain a balanced diet.	Because my busy academic schedule, limited nutritional awareness, and unhealthy eating habits influence my food choices.	Which makes me feel confused about my dietary habits and concerned about my health.
I am a nutrition analyst/college administrator.	I'm trying to analyze students' dietary patterns and recommend healthier eating strategies.	But analyzing large volumes of dietary survey data manually is time-consuming and inefficient.	Because multiple factors such as meal frequency, calorie intake, physical activity, and lifestyle affect students' food choices.	Which makes me feel challenged in making accurate, data-driven decisions to improve students' nutrition.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a college student who wants to maintain a healthy lifestyle.	understand my dietary habits and choose healthier food options.	I find it difficult to identify balanced eating patterns and nutritious meals.	my food choices are influenced by busy schedules, limited awareness, and unhealthy eating habits.	confused about my nutrition and concerned about my overall health.
PS-2	a nutrition analyst/college administrator.	<u>analyze</u> students' food consumption patterns and provide effective dietary recommendations.	<u>analyzing</u> large volumes of student dietary data manually is time-consuming and inefficient.	multiple factors such as calorie intake, meal frequency, lifestyle, and physical activity affect students' health.	challenged in making accurate, data-driven nutrition decisions.

2.2 Empathy Canvas Map



2.3 Brainstorming

Brainstorm & idea prioritization

Use this template in your next brainstorming session so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

1. 10 minutes to prepare
2. 10 minutes to brainstorm

Before you collaborate

A little bit of preparation goes a long way with the online aspect of what you need to do to get going.

1. 5 minutes

2. 5 minutes

3. 5 minutes

4. 5 minutes

5. 5 minutes

6. 5 minutes

7. 5 minutes

8. 5 minutes

9. 5 minutes

10. 5 minutes

Define your problem statement

What problem are you trying to solve? Frame your problem in a way that's clear, concise, and actionable. This will be the focus of your brainstorm.

1. 5 minutes

2. 5 minutes

3. 5 minutes

4. 5 minutes

5. 5 minutes

6. 5 minutes

7. 5 minutes

8. 5 minutes

9. 5 minutes

10. 5 minutes

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

Tip: You can select a sticky note and fit the panel (switch to sketching) to start drawing!

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

Tip: Add color-coded tags to sticky notes to make it easier to find, format, organize, and categorize important ideas as themes within your mural.

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

Tip: Participants can use this Prioritize panel to place all relevant sticky notes on the grid. This facilitator can confirm the grid by using the color-coded legend on the right.

After you collaborate

You can export the mural as an image or pdf to share with members of your company who might find it helpful.

Quick add-ons

Share the mural

Export the mural

Keep moving forward

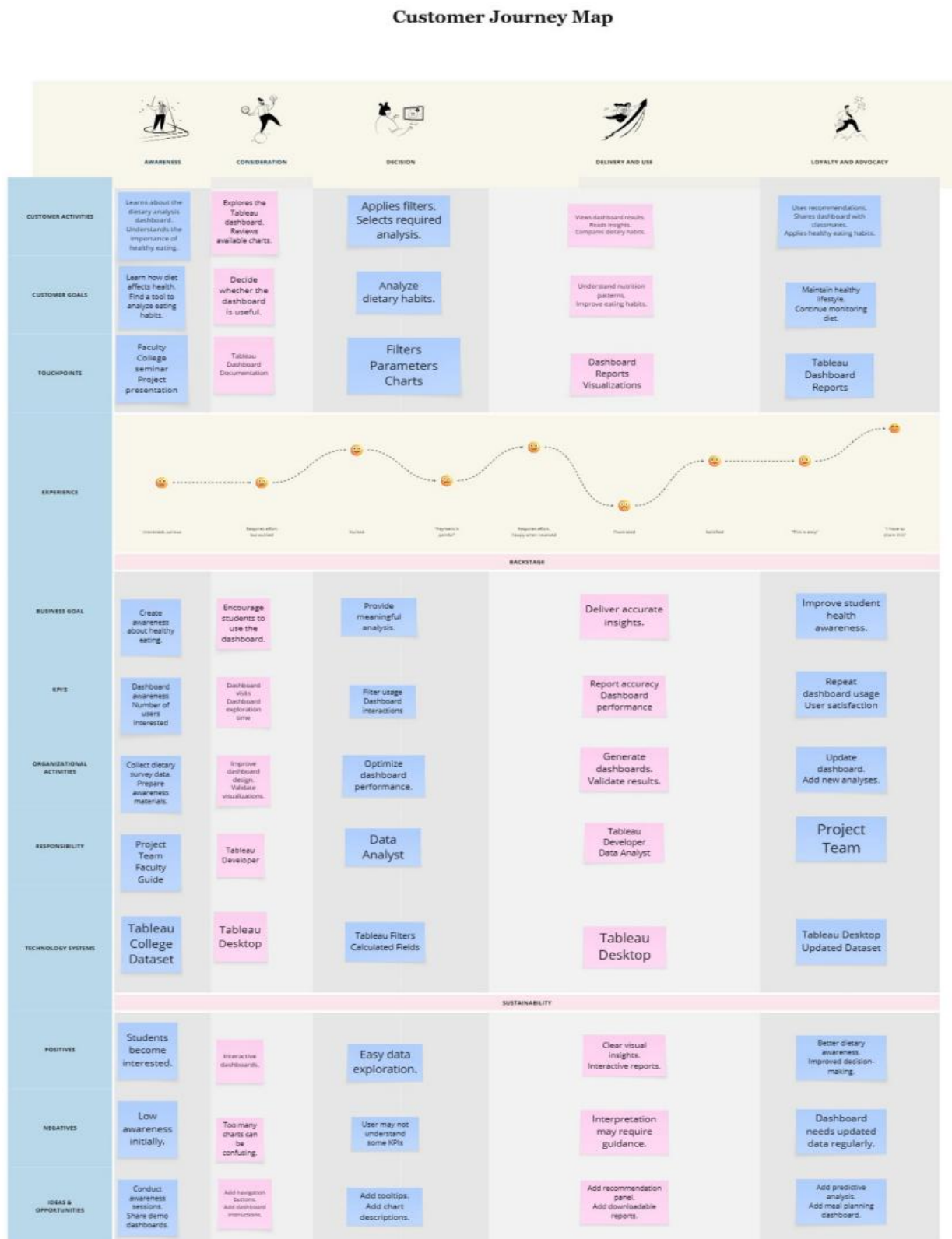
Strategy blueprint

Customer experience journey map

Strengths, weaknesses, opportunities & threats

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

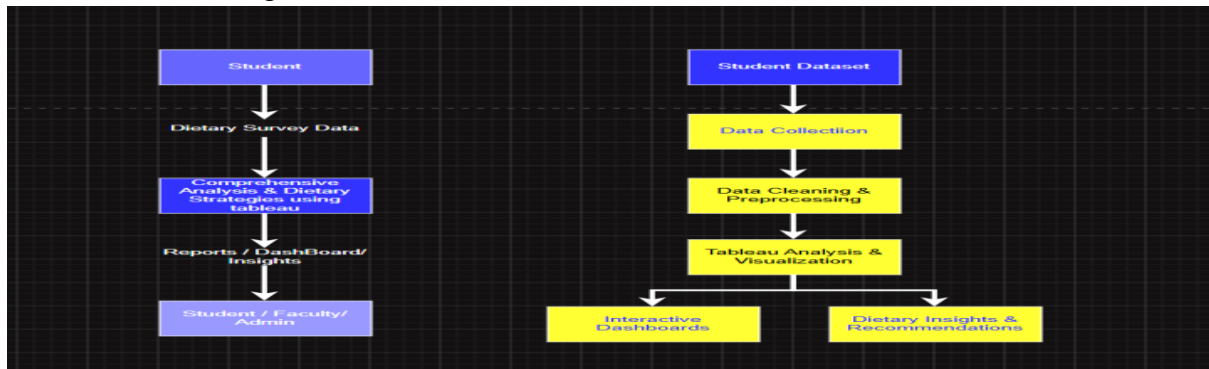
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import dietary dataset from CSV/Excel into Tableau Validate dataset before analysis Handle missing or inconsistent data
FR-2	Data Preprocessing	Clean and transform dataset Create calculated fields Apply filters and data relationships
FR-3	Dashboard Creation	Design interactive Tableau dashboards Create charts and graphs Display KPIs and summary statistics
FR-4	Dietary Analysis	Analyse meal frequency Analyse calorie intake Analyse nutrition and food habits
FR-5	Lifestyle Analysis	Compare diet with physical activity Compare diet with academic performance Analyse BMI and health indicators
FR-6	Reporting & Insights	Generate visual reports Provide dietary insights

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be simple, interactive, and easy to understand.
NFR-2	Security	Dataset should be stored securely with restricted access.
NFR-3	Reliability	Dashboard should display accurate and consistent an
NFR-4	Performance	Dashboard should load quickly and respond efficiently to filters.
NFR-5	Availability	Dashboard should be accessible whenever the Tableau workbook is opened or published.
NFR-6	Scalability	New dietary datasets and visualizations can be added without redesigning the dashboard.

3.3 Data Flow Diagram



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Student	Data Import	USN-1	As a user, I want to import the dietary dataset into Tableau.	Dataset imports successfully without errors.	High	Sprint-1
Student	Data Cleaning	USN-2	As a user, I want to clean missing and inconsistent data before analysis.	Clean dataset is ready for visualization.	High	Sprint-1
Student	Dashboard	USN-3	As a user, I want an interactive dashboard to explore dietary habits.	Dashboard loads with filters and charts.	High	Sprint-1
Student	Nutrition Analysis	USN-4	As a user, I want to analyse calorie intake and meal frequency.	Dashboard displays nutrition metrics correctly.	High	Sprint-1
Student	Lifestyle Analysis	USN-5	As a user, I want to compare diet with physical activity.	Lifestyle comparison charts are available.	Medium	Sprint-2
Student	Health Analysis	USN-6	As a user, I want to view BMI and health indicators.	BMI analysis is displayed accurately.	Medium	Sprint 2
Student	Reporting	USN-7	As a user, I want visual reports for dietary insights.	Reports can be viewed and interpreted easily.	High	Sprint 2
Faculty	Dashboard	USN-8	As a faculty member, I want to monitor students' dietary trends.	Faculty can access summarized dashboards.	Medium	Sprint 2
Nutrition Analyst	Dietary Analysis	USN-9	As an analyst, I want to identify food consumption patterns.	Dashboard provides meaningful trend analysis.	High	Sprint 2
Nutrition Analyst	Recommendation	USN-10	As an analyst, I want to recommend healthier dietary strategies.	Insights support evidence-based recommendations.	High	Sprint 3
Administrator	Data Management	USN-11	As an administrator, I want to update the dataset.	Updated data refreshes the dashboard successfully.	Medium	Sprint 3
Administrator	Dashboard Management	USN-12	As an administrator, I want to maintain dashboard accuracy and availability.	Dashboard remains accessible and up to date.	Medium	Sprint 3

3.4 Technology Stack

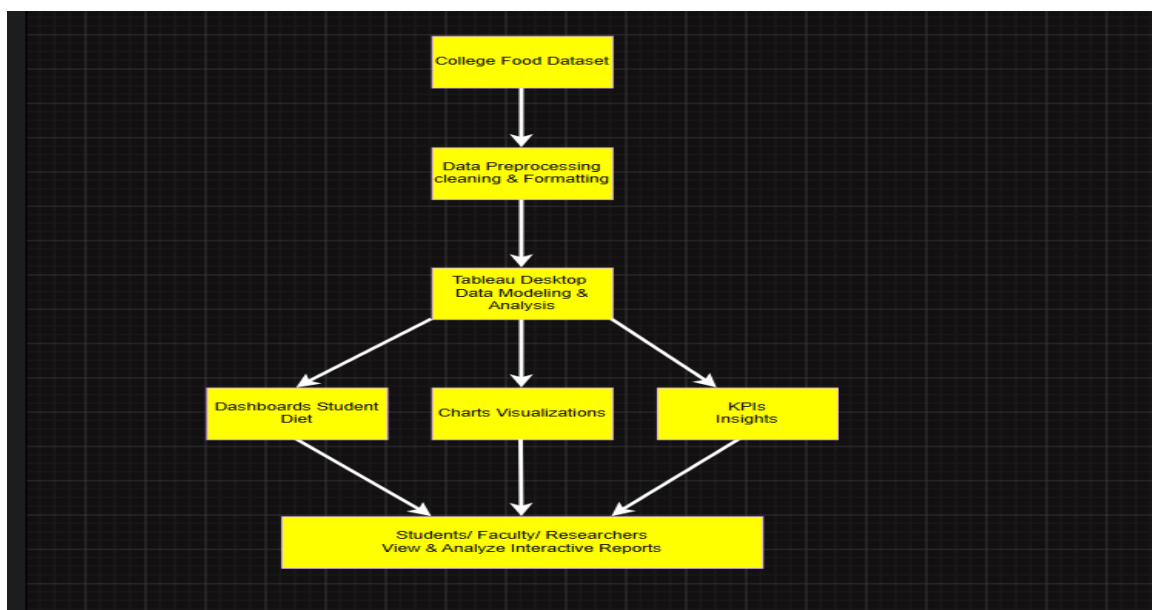


Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for users	Tableau Desktop / Tableau Public
2.	Application Logic-1	Data cleaning and preprocessing	Microsoft Excel
3.	Application Logic-2	Data transformation and calculations	Tableau Calculated Fields
4.	Application Logic-3	Data visualization and dashboard creation	Tableau Desktop
5.	Database	College food choices dataset	CSV Dataset
6.	Cloud Database	Not Applicable	N/A
7.	File Storage	Store dataset and Tableau workbook	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Not Used	N/A
9.	External API-2	Not Used	N/A
10.	Machine Learning Model	Not Applicable	N/A
11.	Infrastructure (Server / Cloud)	Local System	Windows PC with Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Dataset preprocessing and visualization tools	Tableau Public, Microsoft Excel
2.	Security Implementations	Dataset stored locally with restricted access	Local File Permissions
3.	Scalable Architecture	Dashboards can be expanded by adding new datasets and worksheets	Tableau Data Model
4.	Availability	Dashboard accessible whenever Tableau Workbook published	Tableau Desktop / Tableau Public
5.	Performance	Optimized through data cleaning, filters, calculated fields, and efficient visualizations	Tableau Optimization Techniques

4. PROJECT DESIGN

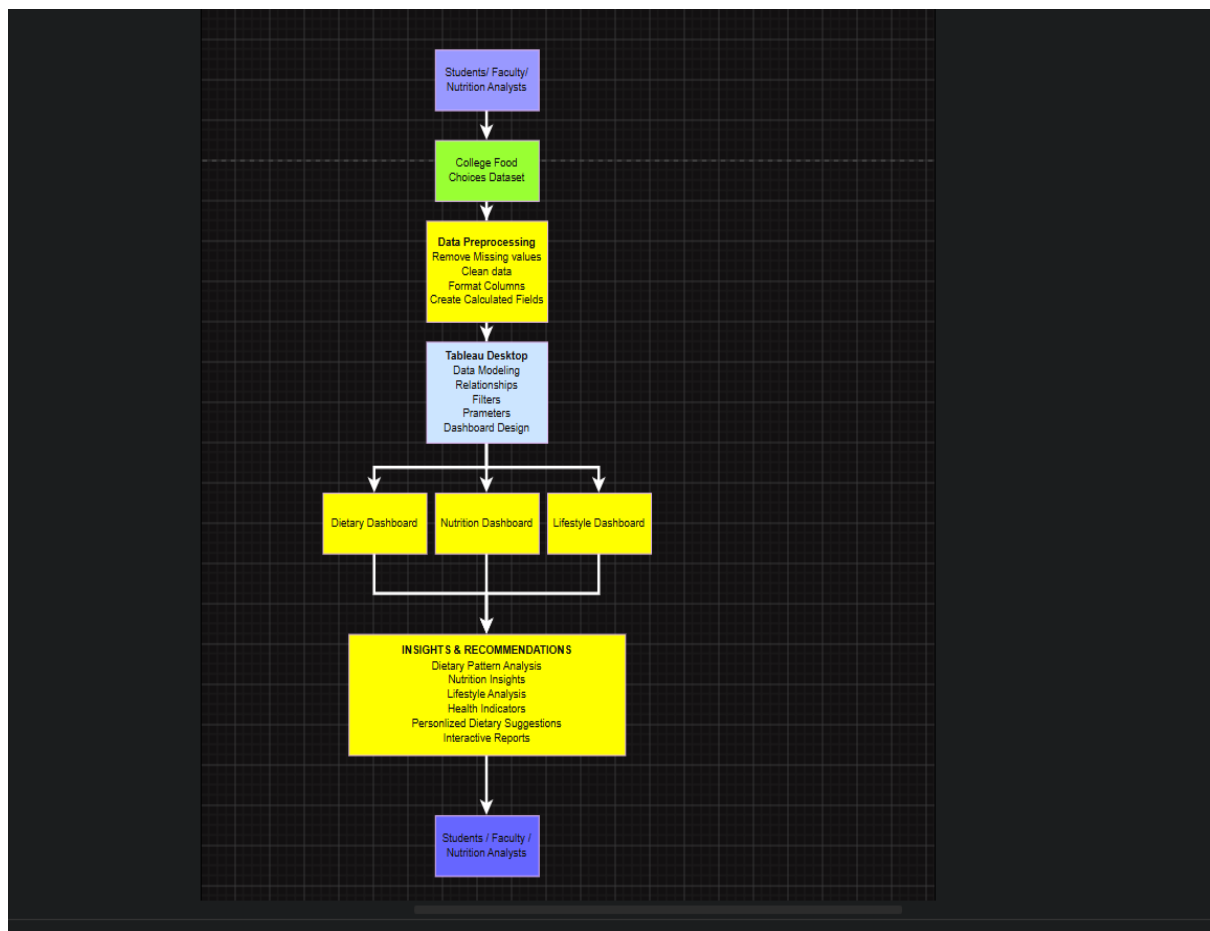
4.1 Problem Student Fit:

Problem-Solution Fit canvas		Purpose / Vision	Version:
Define CS, fit into CL	1. CUSTOMER SEGMENT(S) CS College students University faculty Nutrition Analysts	6. CUSTOMER LIMITATIONS CL EG. BUDGET, DEVICES Busy academic schedule. Budget constraints knowledge. Budget Constraints	5. AVAILABLE SOLUTIONS AS PROS & CONS Nutrition websites Health blogs Diet charts
	Focus on PR, tap into BE, understand RC	2. PROBLEMS / PAINS PR + ITS FREQUENCY Understand daily dietary habits. Analyze nutritional intake. Making informed food choices. Identify lifestyle factors affecting Diet.	9. PROBLEM ROOT / CAUSE RC Lack of awareness about nutrition Busy college schedules Poor meal planning Easy availability of unhealthy food
Identify strong TR & EM		3. TRIGGERS TO ACT TR Health awareness campaigns. Poor eating habits.	10. YOUR SOLUTION SL Analyzes dietary habits. Identifies eating patterns. Generates meaningful insights. Compares lifestyle and dietary factory.
	4. EMOTIONS EM BEFORE / AFTER Before Confused Unaware After Informed Confident	Extract online & offline CH of BE	

4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	College students often face challenges in maintaining healthy dietary habits due to busy academic schedules, limited nutritional awareness, and easy access to unhealthy food options. Analysing dietary data manually is difficult and time-consuming, making it challenging to identify eating patterns and provide meaningful dietary recommendations.
2.	Idea / Solution description	The proposed solution is an interactive Tableau dashboard that analyses college students' dietary habits using survey data. The dashboard presents visual insights into meal frequency, calorie intake, food preferences, lifestyle factors, and nutrition patterns through interactive charts, filters, and KPIs. These insights help students, faculty, and nutrition analysts understand dietary behaviours and support healthier food choices.
3.	Novelty / Uniqueness	• Interactive and user-friendly Tableau dashboards. • Combines multiple dietary and lifestyle factors into a single analytical platform. • Uses visual analytics to simplify complex nutritional data. • Supports data-driven dietary decision-making. • Provides meaningful insights through dynamic filters and visualizations
4.	Social Impact / Customer Satisfaction	• Promotes healthier eating habits among college students. • Increases awareness of balanced nutrition and lifestyle choices. • Helps educational institutions understand students' dietary behaviours.
		• Supports informed health-related decisions using visual analytics. • Enhances student well-being and encourages preventive health practices.
5.	Business Model (Revenue Model)	• Institutional dashboard licensing. • Subscription-based analytics services. • Custom dashboard development for colleges and universities. • Health and nutrition consulting using analytical reports.
6.	Scalability of the Solution	• Integrating larger datasets from multiple institutions. • Publishing dashboards through Tableau Server or Tableau Cloud. • Adding real-time dietary data sources. • Including predictive analytics and machine learning in future versions. • Expanding the analysis to hospitals, fitness centres, and public health organizations.

4.3 Solution Architecture



5. Project Planning & Scheduling:

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Collect the College Food Choices dataset from survey data.	2	High	
Sprint-1	Data Collection	USN-2	Import the dataset into Tableau.	1	High	
Sprint-1	Data Preparation	USN-3	Remove missing values from the dataset.	3	High	
Sprint-1	Data Preparation	USN-4	Create calculated fields and organize data.	3	Medium	
Sprint-1	Data Preparation	USN-5	Handle duplicate and inconsistent records.	3	High	
Sprint-2	Data Visualization	USN-6	Create bar charts for dietary analysis.	2	High	
Sprint-2	Data Visualization	USN-7	Create pie charts for food category distribution.	2	Medium	
Sprint-2	Data Visualization	USN-8	Create line charts for trend analysis.	2	Medium	
Sprint-2	Dashboard Development	USN-9	Develop interactive Tableau dashboards.	5	High	
Sprint-2	Reporting	USN-10	Create dashboard story and final report.	5	High	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	6 Jun 2026	9 Jun 2026	12	9 Jun 2026
Sprint-2	16	6 Days	10 Jun 2026	14 Jun 2026	16	14 Jun 2026
Sprint-3	14	6 Days	13 Jun 2026	18 Jun 2026	14	18 Jun 2026
Sprint-4	10	6 Days	20 Jun 2026	30 Jun 2026	10	30 Jun 2026

$$AV = \frac{\text{Sprint duration}}{\text{velocity}} = \frac{13}{12} = 1.08$$

1.08(Story points per sprint)

6. FUNCTIONAL AND PERFORMANCE TESTING:

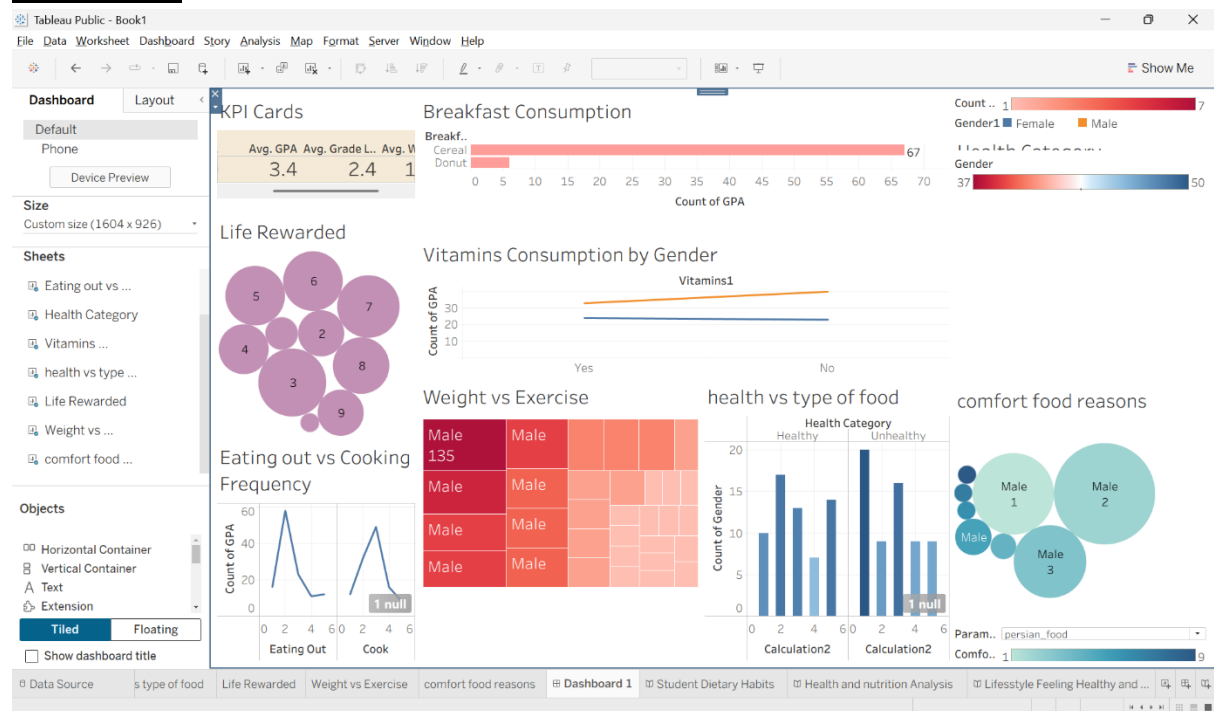
6.1 Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Dataset Name : food_coded.csv Total Records Loaded : 125 Total Columns : 68 Data Source : CSV Data Successfully Loaded into Tableau.
2.	Data Preprocessing	• Connected CSV dataset to Tableau. • Verified data types. • Created aliases for coded values (Gender, Breakfast). • Created calculated fields for Health Category. • Created parameters for food category selection. • No duplicate records found. • Dataset was already pre-cleaned.
3.	Utilization of Filters	Gender Health Category Parameter 2 (Vitamins) Parameter 3 (Greek Food) Breakfast Exercise
4.	Calculation fields Used	3 calculation fields are used: Health category, vitamins parameter calculation, Type of food
5.	Dashboard design	No of Visualizations / Graphs - 9
6	Story Design	No of Visualizations / Graphs - 7

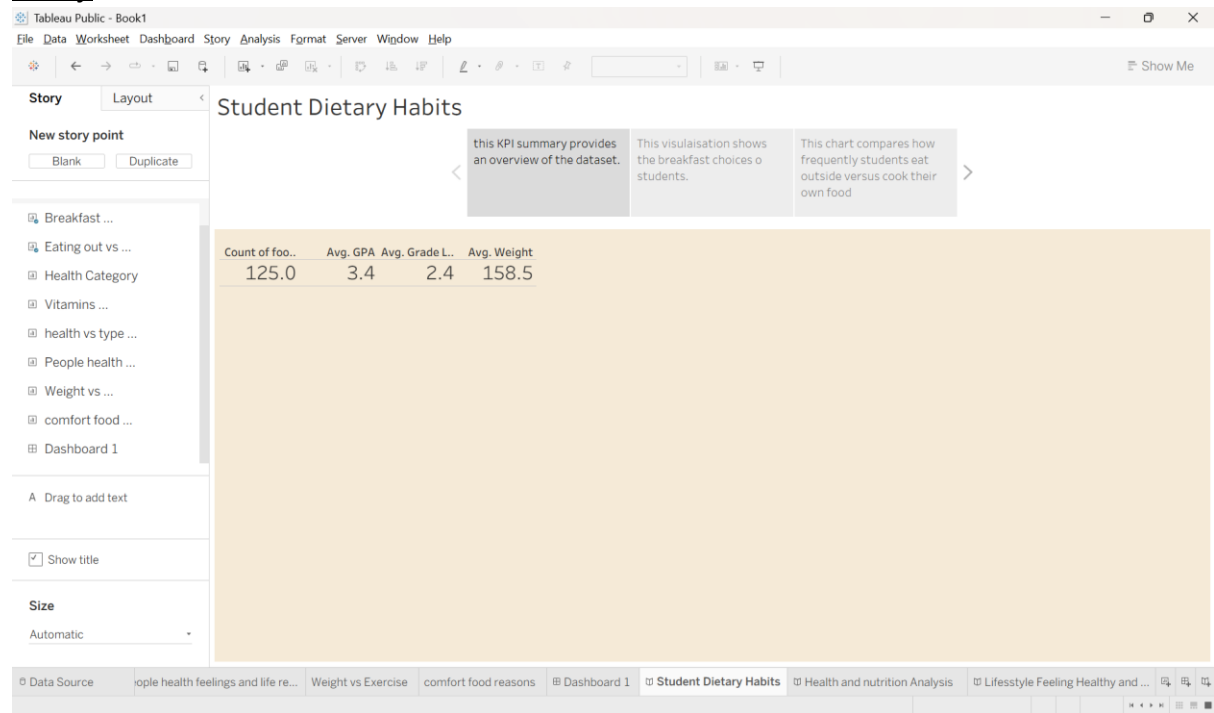
7. RESULTS:

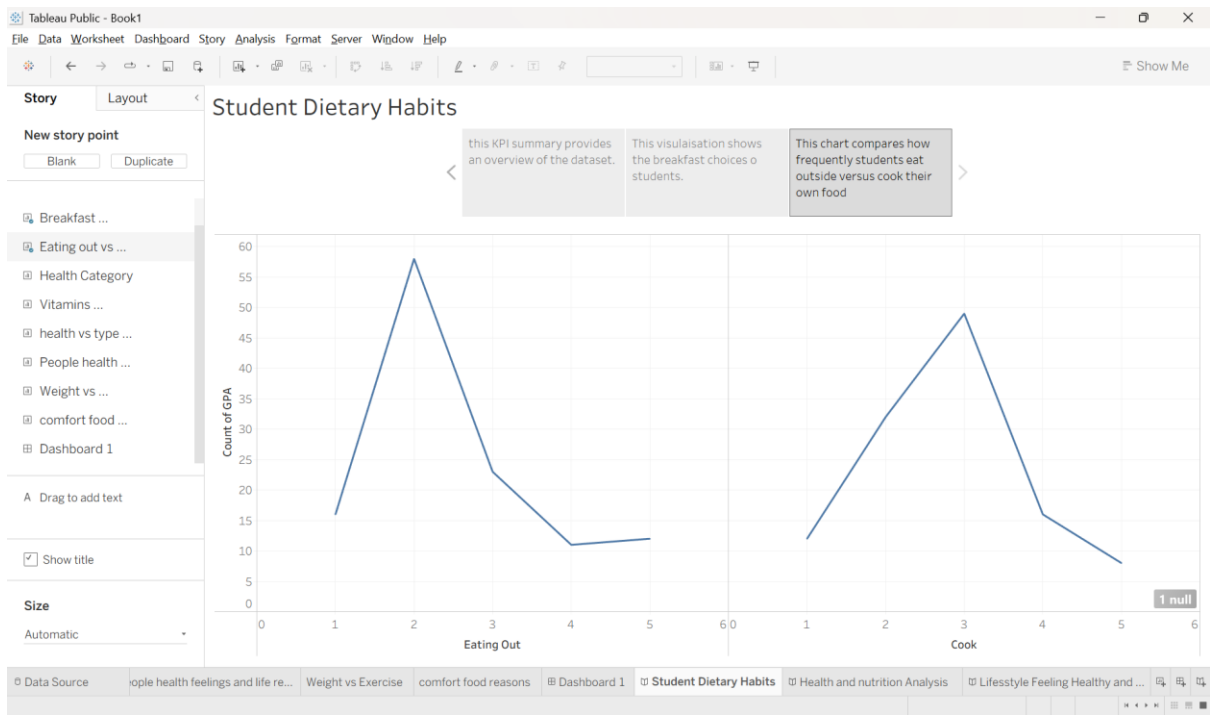
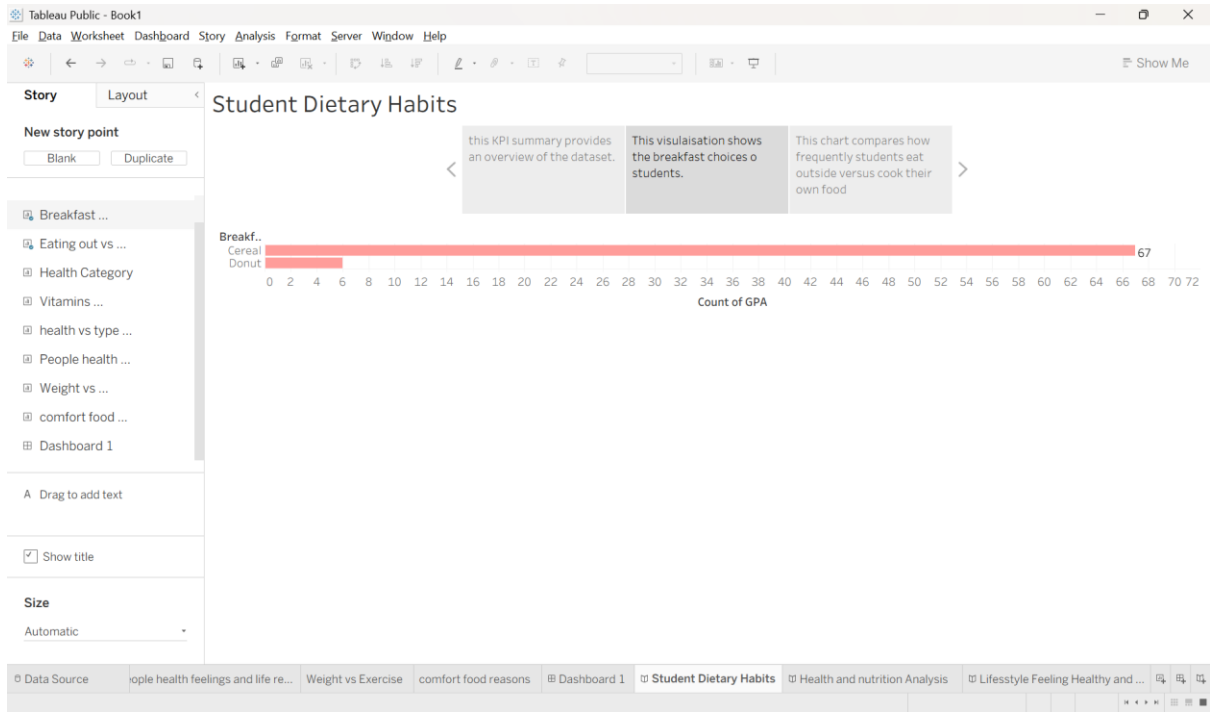
7.1 Output Screenshots

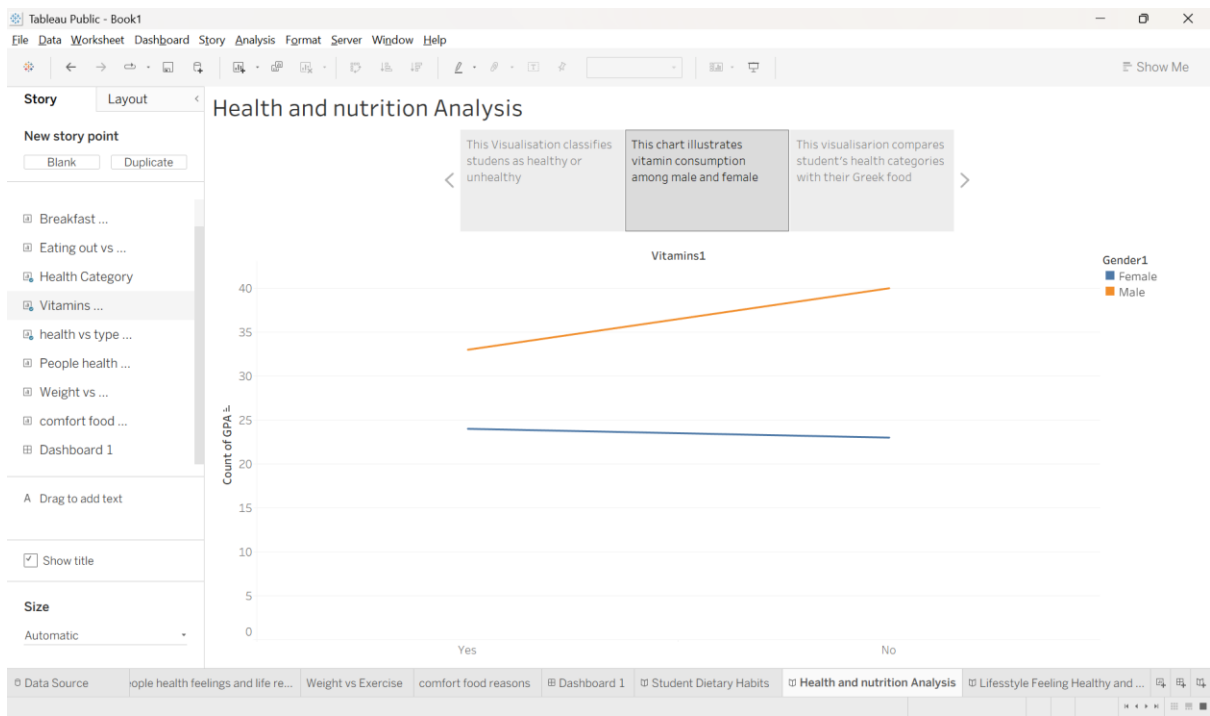
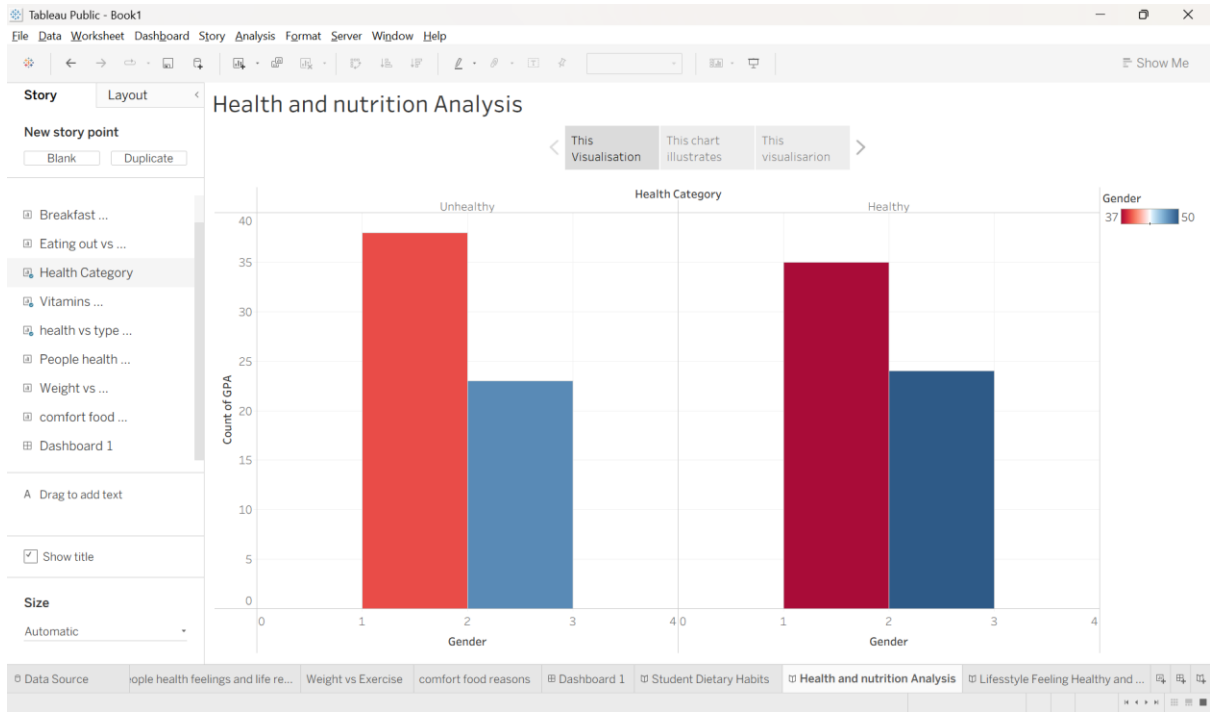
Dashboard

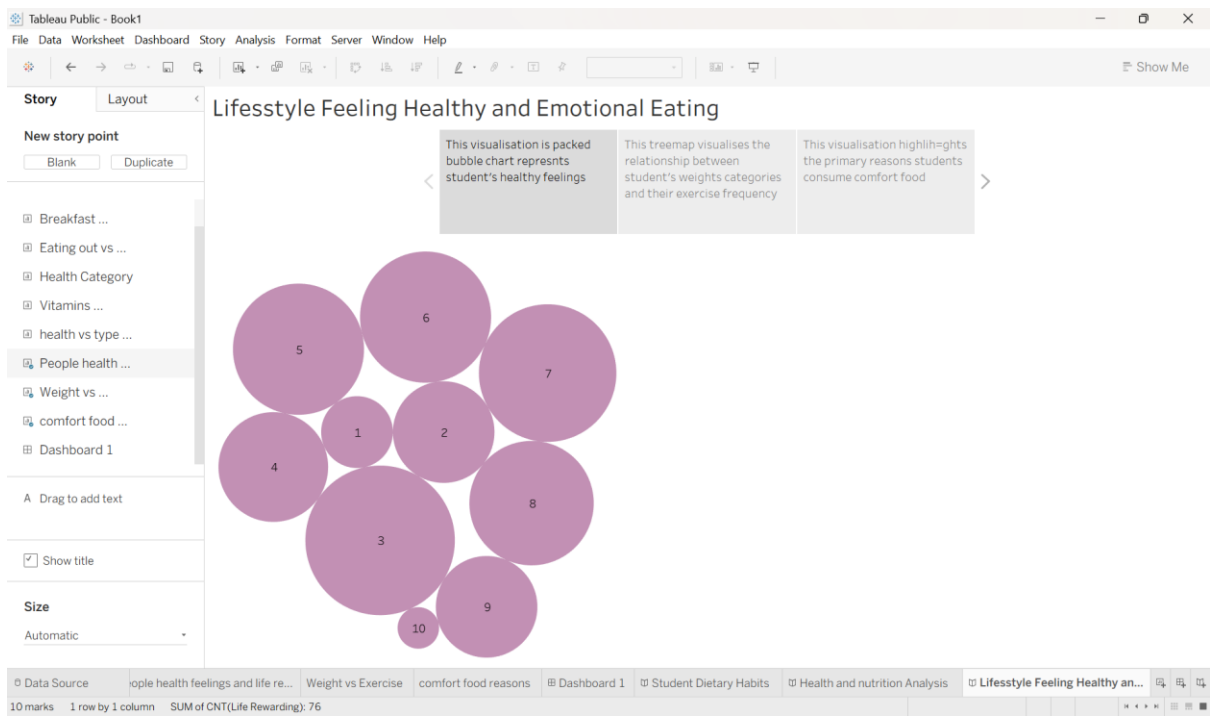
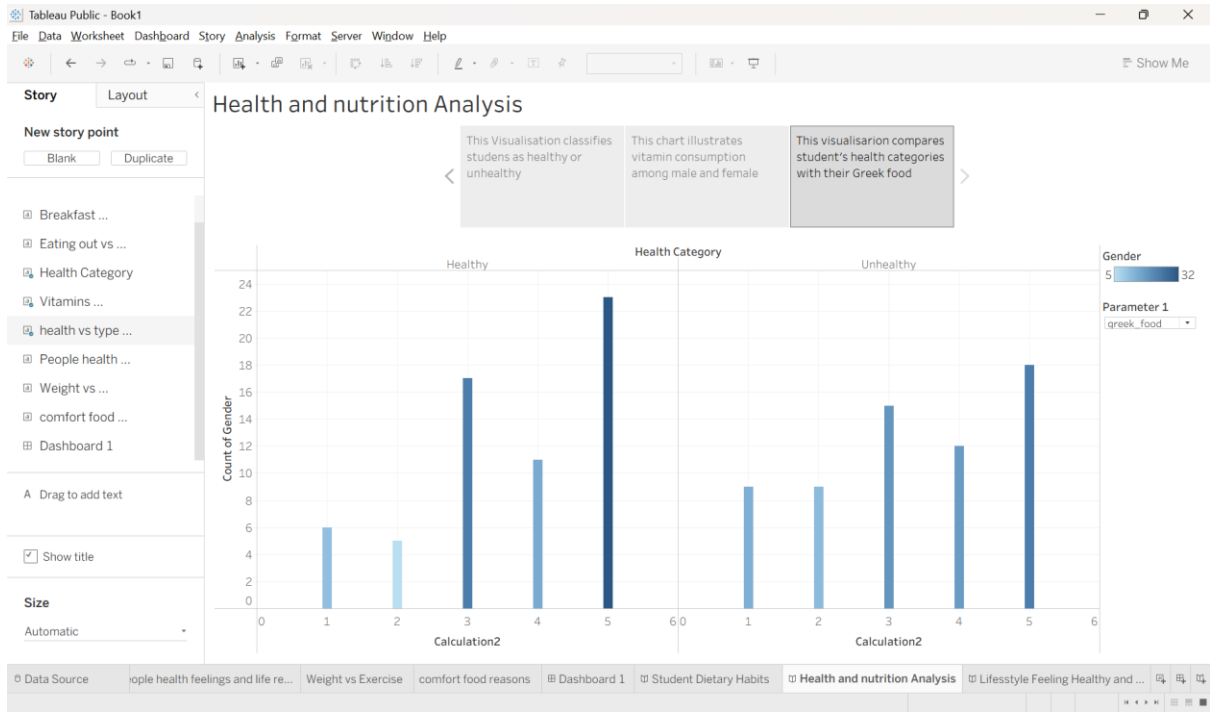


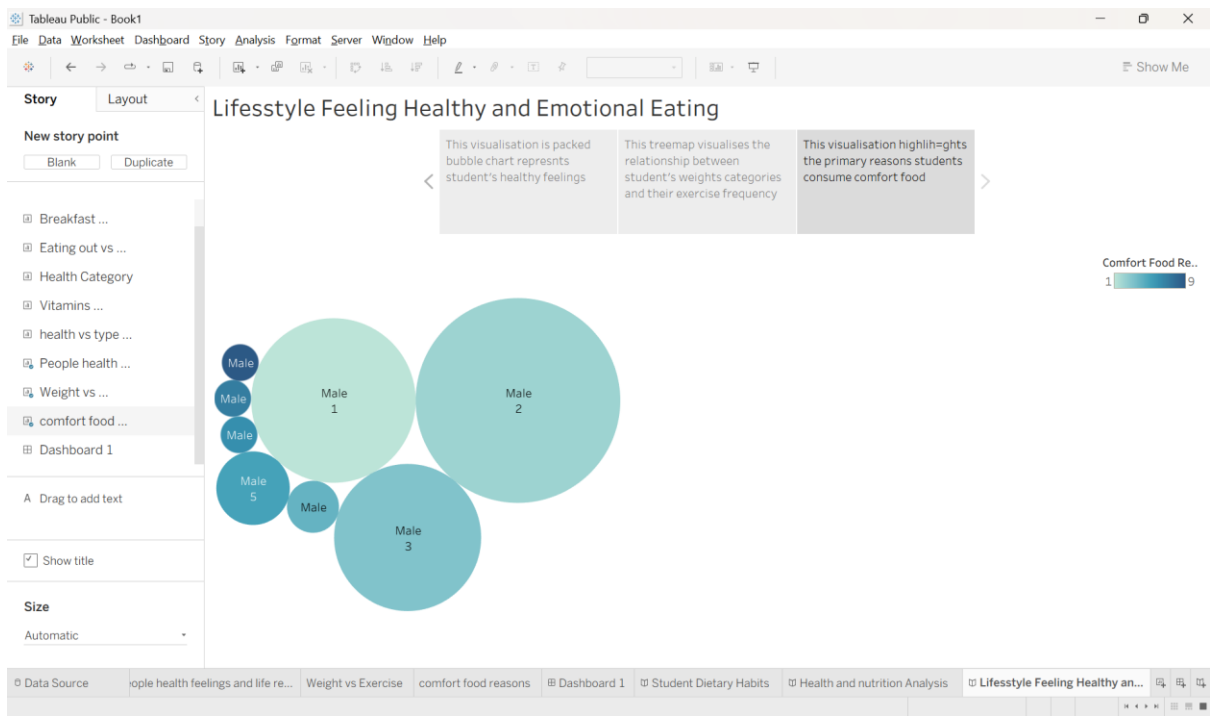
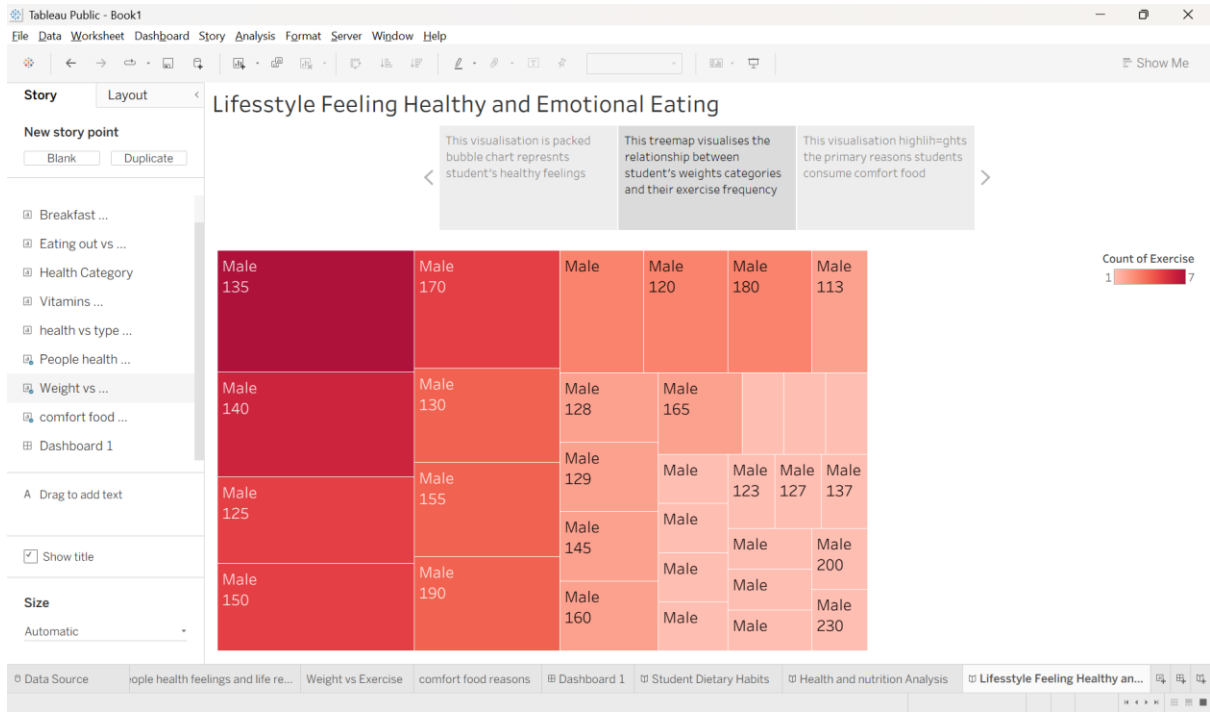
Story











8. ADVANTAGES AND DISADVANTAGES:

Advantages

- Provides clear visualization of students' dietary habits.
- Helps identify healthy and unhealthy eating patterns.
- Interactive dashboards make data easy to understand.
- Supports data-driven decision-making for educational institutions.
- Enables quick comparison of food preferences across different student groups.
- Uses filters and parameters for dynamic data exploration.
- Simplifies complex survey data into meaningful insights.
- Helps identify relationships between diet, exercise, and health.
- Improves awareness about healthy lifestyle choices.
- User-friendly interface with interactive dashboards and story points.
- Easy to update with new datasets for future analysis.
- Supports effective presentation of analytical results.

Disadvantages

- Analysis is limited to the available survey dataset.
- Results depend on the accuracy of self-reported responses.
- Does not provide real-time data analysis.
- Limited sample size may not represent all college students.
- Does not predict future dietary behaviour.
- External factors such as income and location are not deeply analysed.
- Requires Tableau knowledge to modify or extend the dashboard.
- Dashboard performance may decrease with very large datasets.
- Findings are specific to the collected dataset and cannot be generalized universally.
- Additional datasets are required for more comprehensive analysis.

9. Conclusion:

The Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study successfully analyzed the dietary habits, health perceptions, and lifestyle patterns of college students using interactive Tableau visualizations. The project transformed raw survey data into

meaningful insights through KPI cards, charts, dashboards, and story points, making it easier to understand students' food choices and their impact on overall well-being.

The analysis revealed important patterns such as breakfast consumption habits, eating-out frequency, comfort food preferences, exercise behaviour, and the relationship between health perception and dietary choices. These visualizations help identify factors influencing students' nutritional habits and support informed decision-making for promoting healthier lifestyles.

The interactive dashboard and story provide a user-friendly platform for exploring the data, allowing users to compare different aspects of student health and food consumption efficiently. Overall, the project demonstrates how Tableau can be effectively used as a Business Intelligence and data visualization tool to convert complex datasets into actionable insights. The findings can assist educational institutions, nutritionists, and student wellness programs in developing strategies to encourage healthier eating habits, improve student well-being, and support better academic performance.

10. FUTURE SCOPE:

Some possible future enhancements include:

- Integrating real-time food and nutrition data from mobile or web applications.
- Expanding the dataset to include students from multiple colleges and universities for broader analysis.
- Applying Machine Learning algorithms to predict students' dietary habits and health outcomes.
- Developing personalized diet recommendations based on individual food preferences and health conditions.
- Integrating fitness and health tracking data such as physical activity, sleep patterns, and calorie expenditure.
- Creating mobile-friendly and responsive dashboards for easy access on smartphones and tablets.
- Adding advanced Tableau features such as forecasting, trend analysis, and predictive analytics.
- Publishing the dashboard on Tableau Public or a web application for wider accessibility.
- Including additional health factors such as BMI, medical history, stress levels, and nutritional intake for more comprehensive analysis.

- Enabling automated data updates to keep the dashboard current without manual intervention.

11. APPENDIX:

Dataset Link:

https://drive.google.com/drive/folders/19h0g1luzEblf6FrI9RY7eAAS4sTtOu_P

GitHub & Project Demo Links:

GitHub:

<https://github.com/SaiHemanth1106/Comprehensive-Analysis-and-Dietary-Strategies-with-Tableau-A-College-Food-Choices-Case-Study>

Demo Link:

<https://drive.google.com/drive/folders/11uwsUJLHrVy7cWHHteDYOG3e4jkY0fX5>